A Note on Deformation Theory

May 2026

I wanted to make a short note providing intuition and clarifying a few facts about first order deformations of closed subschemes.

1 Hilbert Scheme

The functor of points perspective to algebraic geometry is to regard an S-scheme as a certain type of functor

$$(\mathrm{Sch}/S)^{\mathrm{op}} \to \mathrm{Set},$$

and morphisms of schemes as any natural transformation. Let X, M be two such functors, and assume X is representable (at least). Then, the Yoneda lemma tells us

$$\mathrm{Hom}(X, M) \cong M(X).$$

In particular, we consider the case when M is a *moduli space*. This is a somewhat vague notion (in the sense that every scheme is the moduli space of its own points), but we mean that there is a type of object that you want to parameterize, and you have a definition for a *family* of those objects parameterized by an S-scheme T. In this case, the functor you want M to represent is

$$T \mapsto \{\text{families of this type of object parameterized by } T\}.$$

We give an example. Fix a k-scheme X and a $d \in \mathbb{Z}_{\geq 1}$.

Definition 1.1. A **family of d points on X over (or parameterized by)** T is a diagram

$$\begin{tikzcd} Z \arrow[r, hook] \arrow[dr] & X \times_k T \arrow[d] \\ & T \end{tikzcd}$$

such that the diagonal arrow is flat, the right arrow is a closed embedding, and for each $t \in T$, the base changed diagram

$$\begin{tikzcd} Z_t \arrow[r, hook] \arrow[dr] & X \times_k \operatorname{Spec} k(t) \arrow[d] \\ & \operatorname{Spec} k(t) \end{tikzcd}$$

is a closed embedding of a length d subscheme $Z_t \hookrightarrow X_{k(t)}$ (e.g., the inclusion of d distinct k points when $k(t) = k$).

Definition 1.2. The **Hilbert functor of d points of X**, denoted $X^{[d]}$, is the functor

$$\begin{aligned}(\mathrm{Sch}/k)^{\mathrm{op}} &\longrightarrow \mathrm{Set}\\ T &\longmapsto \{\text{families of } d \text{ points on } X \text{ over } T\}/\sim,\end{aligned}$$

where the equivalence relation $\sim$ is an isomorphism of diagrams, which amounts to an isomorphic closed embedding.

Fact. This functor is representable, so we can instead say the **Hilbert scheme of d points of X**, and abuse notation and still write $X^{[d]}$.

There are more general Hilbert schemes depending on a polynomial $P \in \mathbb{Q}[t]$, denoted $\mathrm{Hilb}^P X$, where we require additionally that X is quasiprojective, and then a family over T is a Z with $t \in T$ fibers being closed subschemes of $X_{k(t)}$ with prescribed Hilbert polynomial P.

Note that a k-point of X is just a closed subcheme $Z \hookrightarrow X$ with Hilbert polynomial P. Setting P to be the constant polynomial d, we recover $X^{[d]}$ (at least when X is quasiprojective).

2 Deformations

It is not very obvious that $X^{[d]}$ is always representable, but the proof does embed $X^{[d]}$ in a Grassmannian (which then embeds in a projective space), but it would be trully horrible to reason about $X^{[d]}$ by using its defining equations in some ginormous ambient space.

Instead, we use the moduli functor description to reason about the geometry. Let $D = \operatorname{Spec} k[\varepsilon]/\varepsilon^2$ be the dual numbers. Recall the following bijection:

Proposition 2.1. Let M be a finite type k-scheme with k-point x. Denote $\mathfrak{m} \leq \mathcal{O}_{M,x}$ the maximal ideal of the stalk at x. Then,

$$T_x M = \mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k) \leftrightarrow \{\varphi \in \mathrm{Hom}_{\operatorname{Spec} k}(\operatorname{Spec} D, X) \mid \varphi((\varepsilon)) = x\}.$$

If we identify $x \in M$ with the map $x : \operatorname{Spec} k \to M$ and denote $i : \operatorname{Spec} k \to \operatorname{Spec} D$ the inclusion of the closed point, we can even rewrite this bijection as

$$\{\varphi \in \mathrm{Hom}_{\operatorname{Spec} k}(\operatorname{Spec} D, X) \mid \varphi \circ i = x\}.$$

The advantage here is that when M has a nice moduli description, we can interpret this entirely moduli-theoretically. For example, let M be $\mathrm{Hilb}^P X$. When we unpack the definition of a tangent vector of a $[Z] \in \mathrm{Hilb}^P X$, we instead call it a first-order deformation, but this is just a change in language.

Definition 2.2. Denote $X' = X \times_k \operatorname{Spec} D$. A **first-order deformation** of $Z \subseteq X$ is a diagram

$$\begin{tikzcd} Z' \arrow[r, hook] \arrow[dr] & X' \arrow[d] \\ & \operatorname{Spec} D \end{tikzcd}$$

with the diagonal arrow flat, such that the diagram base changes to our given $Z \hookrightarrow X$.

Remark 2.3. When we consider specifically $P = 1$ to study the tangent space of $X^{[1]} = X$, we might think we are exactly recovering the proposition as a special case. However, we are actually doing something slightly different!

Note that in the proposition, we totally fix $\operatorname{Spec} D$—the right hand side is *not* the set of isomorphism classes of closed embeddings, for two reasons:

- We are completely fixing $\operatorname{Spec} D$ up to equality, not isomorphism.
- Not all maps $\operatorname{Spec} D \to X$ are closed embeddings, since we do not need to be surjective on the level of rings by doing a factorization

  $$\operatorname{Spec} D \to \operatorname{Spec} k \to X$$

  (where the first map is very much not an embedding, you kill nilpotents).

As for this second point, $\operatorname{Spec} D \to X$ will be a closed embedding unless it represents the 0 tangent vector. But, we do get a morphism $\operatorname{Spec} D \to X'$ which is *always* a closed embedding.

Furthermore, in general a closed embedding $\operatorname{Spec} D \subseteq X'$ (where we only remember $\operatorname{Spec} D$ as a closed subscheme) is *more* data than a closed subscheme $\operatorname{Spec} D \subseteq X$. Passing to the reduction will exactly correspond to quotienting

$$(T_x X \setminus \{0\})/k^*$$

getting the projectivized tangent space.

For example, for $X = \operatorname{Spec} k[x, y]$, we have several first order deformations of $0 \in X$ which are

$$Z'_a = V(x - a\varepsilon, y) \subseteq X',$$

and when $a \neq 0$ (and maybe $\operatorname{char} k = 0$), the image of all of these $Z'_a \to X' \to X$ are just $V(x^2, y)$.

Remark 2.4. In one way or another, however, we do get that

$$\{\text{1st-order deformations of } \{x\} \subseteq X\} \leftrightarrow \operatorname{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k),$$

but our remark makes this slightly awkward, and we would like generalizations. This leads us to the following.

Proposition 2.5.

$$\{\text{1st-order deformations of } Z \subseteq X\} \leftrightarrow \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{I}_Z, \mathcal{O}_Z) \cong H^0(Z, \mathcal{N}_{Z/X}).$$

I want to provide a proof of this proposition that is very well geometrically motivated, so I need to make some comments.

Remark 2.6. Sections of the normal bundle should be thought of as like families of derivations. Recall that a k-derivation on $\mathcal{O}_{X,x}$ is determined by its value on $\mathfrak{m}$, and corresponds to a $\mathcal{O}_{X,x}$ morphism

$$\mathfrak{m} \to \mathcal{O}_{X,x}/\mathfrak{m}.$$

A prototypical example is when $X = \operatorname{Spec} k[x,y]$, $\mathfrak{m} = (x,y)$, and the derivation corresponding to the e_1-tangent vector, which is the morphism

$$\begin{aligned} \mathfrak{m} &\longrightarrow \mathcal{O}_{X,x}/\mathfrak{m} \\ f &\longmapsto \frac{\partial f}{\partial x}(0), \end{aligned}$$

i.e., the derivation picks up the linear term when f is power series expanded in x.

In general, a section of the normal bundle is a similar type of object, which takes sections f of $\mathcal{I}_Z$, expands them in a "direction", and picks out a linear term. Picking a $z \in Z$, we can specialize a section $\mathcal{I}_Z \to \mathcal{O}_X/\mathcal{I}_Z$ to a tangent vector $\mathfrak{m}_z \to \mathcal{O}_X/\mathfrak{m}_z$ by how?

TODO. want to motivate the lifting along $I' \to I$, and how this amounts to Taylor-expanding in the "direction" of the deformation.

3 Regularity of Certain Hilbert Schemes

Using this technology, we can prove the regularity of certain Hilbert schemes.

Roughly, for a surface S, $S^{[d]}$ is smooth because

1. $S^{[d]}$ is connected (induction, using projections from the universal family).
2. calculate $\dim \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{I}_Z, \mathcal{O}_Z) = 2d$.

Proposition 3.1. $\dim \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{I}_Z, \mathcal{O}_Z) = 2d$.

Proof. First, calculate

$$\operatorname{Hom}(\mathcal{I}_Z, \mathcal{O}_Z) \cong \operatorname{Ext}^1(\mathcal{O}_Z, \mathcal{O}_Z)$$

(in analogy to seeing tangent vectors as extensions of skyscraper sheaves). One can see this by hitting

$$0 \to \mathcal{I}_Z \to \mathcal{O}_X \to \mathcal{O}_Z \to 0$$

with $\operatorname{Ext}^*(-, \mathcal{O}_Z)$, to get

$$\operatorname{Hom}(\mathcal{O}_Z, \mathcal{O}_Z) \xrightarrow{\rho^*} \operatorname{Hom}(\mathcal{O}_X, \mathcal{O}_Z) \xrightarrow{i^*} \operatorname{Hom}(\mathcal{I}_Z, \mathcal{O}_Z) \to \operatorname{Ext}^1(\mathcal{O}_Z, \mathcal{O}_Z) \to \operatorname{Ext}^1(\mathcal{O}_S, \mathcal{O}_Z),$$

but note that $\operatorname{Ext}^1(\mathcal{O}_S, \mathcal{O}_Z) = H^1(Z, \mathcal{O}_Z) = 0$ and ρ^* is an isomorphism which implies $i^* = 0$. Thus, we get

$$\operatorname{Hom}(\mathcal{I}_Z, \mathcal{O}_Z) \cong \operatorname{Ext}^1(\mathcal{O}_Z, \mathcal{O}_Z)$$

as desired.

We can rewrite

$$\mathrm{Ext}^i(\mathcal{O}_Z, \mathcal{O}_Z) = \mathrm{Hom}_{D^b(X)}(\mathcal{O}_X, \mathcal{O}_Z \otimes \mathcal{O}_Z^\vee[i]),$$

where we mean the derived dual, so we are computing

$$\mathcal{H}^1(R\Gamma(\mathcal{O}_Z \otimes \mathcal{O}_Z^\vee)).$$

Note that

$$\mathrm{Ext}^0(\mathcal{O}_Z, \mathcal{O}_Z) = \mathrm{Hom}(\mathcal{O}_Z, \mathcal{O}_Z) \cong \Gamma(Z, \mathcal{O}_Z)$$

is n-dimensional, and

$$\mathrm{Ext}^2_{\mathcal{O}_X}(\mathcal{O}_Z, \mathcal{O}_Z) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_Z, \mathcal{O}_Z \otimes \omega_X)^\vee \cong \Gamma(Z, \mathcal{O}_Z \otimes \omega_Z)$$

is also n-dimensional, so that

$$\chi(R\Gamma(\mathcal{O}_Z \otimes \mathcal{O}_Z^\vee)) = n - \dim \mathcal{H}^1(R\Gamma(\mathcal{O}_Z \otimes \mathcal{O}_Z^\vee)) + n = 2n - \dim \mathcal{H}^1(R\Gamma(\mathcal{O}_Z \otimes \mathcal{O}_Z^\vee)).$$

Therefore, we can calculate our dimension using Hirzebruch Riemann Roch:

$$2n - \dim \mathcal{H}^1(R\Gamma(\mathcal{O}_Z \otimes \mathcal{O}_Z^\vee)) = \int_X \mathrm{ch}(\mathcal{O}_Z \otimes \mathcal{O}_Z^\vee)\,\mathrm{td}(T_X),$$

where this integral is actually 0, since the Chern character of $\mathcal{O}_Z$ only exists in top degree (say by Grothendieck Riemann Roch, being a pushforward of along 0 dimensional closed embedding), so the product

$$\mathrm{ch}(\mathcal{O}_Z)\,\mathrm{ch}(\mathcal{O}_Z)^\vee\,\mathrm{td}(T_X)$$

has no terms in the top degree of X. Thus,

$$\dim \mathrm{Ext}^1(\mathcal{O}_Z, \mathcal{O}_Z) = \dim \mathcal{H}^1(R\Gamma(\mathcal{O}_Z \otimes \mathcal{O}_Z^\vee)) = 2n.$$

□

3.1 Nested Hilbert Schemes

Define $S^{[d_1, \cdots, d_\ell]}$ by an incidence correspondence on the product

$$S^{[d_1]} \times \cdots \times S^{[d_\ell]}.$$

In particular, for

$$S^{[d-1,d]} \subseteq S^{[d-1]} \times S^{[d]},$$

the tangent space to a $[Z \subseteq W]$ is a subspace of

$$\mathrm{Hom}(\mathcal{I}_Z, \mathcal{O}_Z) \times \mathrm{Hom}(\mathcal{I}_W, \mathcal{O}_W).$$

To see this subspace, note that $S^{[d-1,d]}$ is cut out by the relation $[Z] \subseteq [W]$. Denoting $i : \mathcal{I}_W \to \mathcal{I}_Z$ the inclusion of ideals and $\rho : \mathcal{O}_W \twoheadrightarrow \mathcal{O}_Z$ the restriction, we claim

$$T_{[Z \subseteq W]} S^{[d-1,d]} = \{(\varphi, \psi) \in \mathrm{Hom}(\mathcal{I}_Z, \mathcal{O}_Z) \times \mathrm{Hom}(\mathcal{I}_W, \mathcal{O}_W) \mid \varphi \circ i = \rho \circ \psi\}.$$

We try to similarly calculate this dimension. It really should have the same dimension as the reduced locus, where we have the flexibility to choose choose d points in an n-dimensional space with one "remaining" point marked. So, we can construct the reduced locus as

$$(S^{\times d} \setminus \{\text{repeated points}\})/S_{d-1},$$

where S_{d-1} acts on the last few coordinates. This has dimension $2d$. (You can also see that $S^{[d-1,d]}$ fibers over $S^{[d]}$ generically finite, so $\dim S^{[d-1,d]} = \dim S^{[d]} = 2d$).

Now, to repeat Fogarty's argument to show $S^{[d-1,d]}$ is smooth, we proceed in two steps:

- Show $S^{[d-1,d]}$ is connected.
- Calculate the tangent space dimension at every closed point of $S^{[d-1,d]}$.

For the first point, we show $S^{[d-1,d]}$ is connected by studying the projection

$$p : S^{[d-1,d]} \to S^{[d]}.$$

We want to bootstrap the connectedness of $S^{[d]}$ from Fogarty to get the connectedness of our space. We need to assume S is proper here I believe.

Proof. Suppose, to the contrary, that $S^{[d-1,d]}$ has a separation $E \sqcup F$ (by closed sets), and we want to argue that $p(E), p(F)$ is a separation of $S^{[d]}$. Since S is proper, so is $S^{[d-1,d]}$ (by a usual valuative criterion argument by existence of flat limits). So, $p(E), p(F)$ are both closed in $S^{[d]}$, so we only need to show they are disjoint and proper.

To see disjointness, notice that if $W \in p(E) \cap p(f)$, then $p^{-1}(W)$ meets both E and F. However, $p^{-1}(W)$ is connected... why?. □

Note that (assuming S is projective) $S^{[d-1,d]}$ is projective (by the valuative criterion) so that p is a closed morphism. if , then $p(E), p(F)$ are close

Now, p is both closed and surjective, so it i

using the following lemma.

Lemma 3.2. I think I need f to be a quotient map, and I'm not sure when this happens in algebraic geometry (for example I'm not sure in my setting). I might need a more algebraic proof. Let $f : X \to Y$ be continuous, Y connected, and all fibers $f^{-1}(y)$ connected nonempty. (arguably the empty set isn't connected, because some conventions regard $\emptyset$ as always a proper subset.)

Proof. Suppose X is not connected, so has a separation $U \sqcup V$. Since $f^{-1}(y)$ is connected, it must lie entirely in U or V, since otherwise $f^{-1}(y) \cap U, f^{-1}(y) \cap V$ provides a separation.

This (using nonemptyness of fibers) implies that $f^{-1}(f(U)) = U$, $f^{-1}(f(V)) = V$. We claim that $f(U), f(V)$ form a separation of Y, but we first need $f(U), f(V)$ to be open. □

Since $S^{[d]}$ is connected by Fogarty, we just have to argue that the fibers are all nonempty and connected.